

Spatial TrackFormer for Cross-Agent Object Association

Progress Report: Embedding-Aware Cooperative LiDAR Perception

Dhia

July 2026

TrackFormer [1] is adapted from temporal tracking to same-timestamp cross-agent association in OPV2V.

What We Built

Detector and data

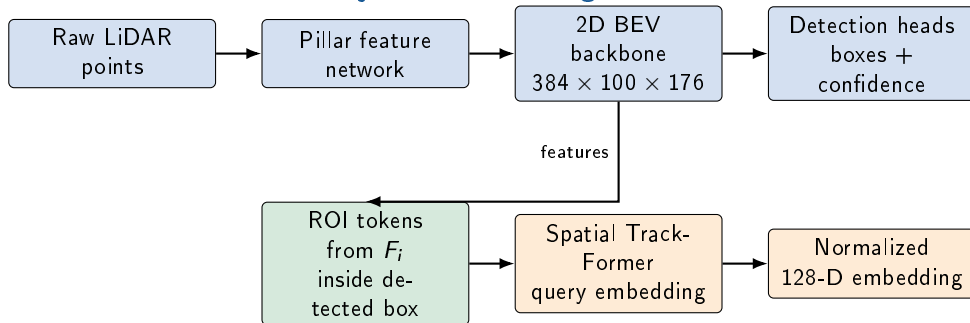
- ▶ OpenCOOD PointPillars runs on the target hardware.
- ▶ Clean late fusion reproduces $AP@0.7 \approx 0.78$.
- ▶ We cache detections, identities, confidence, box geometry, and every vehicle-covered BEV feature cell.

Association objective

- ▶ Given a source-agent detection and an ego-agent detection at the same timestamp, decide whether they are the same physical vehicle.
- ▶ Transform box geometry into the ego frame before comparison.
- ▶ Output a normalized 128-D association embedding per proposal.

Key correction: the original MLP read only one BEV feature cell; the final model reads all cells covered by the vehicle box.

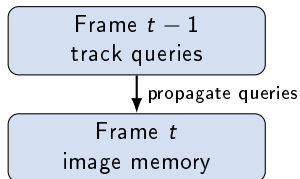
From LiDAR Points to an Object Embedding



- ▶ **Pillar feature network:** a point-wise MLP and max pooling convert all points in one vertical XY cell into one learned pillar descriptor.
- ▶ **2D BEV backbone:** a CNN scatters these descriptors onto the ground plane and combines neighbouring pillars into contextual features.
- ▶ 384 is the number of learned channels, *not* the number or size of pillars; 100×176 is the final spatial grid after the backbone's strides, starting from the configured LiDAR range and 0.4 m voxel grid.
- ▶ **Detection heads** are small convolutional layers that predict a vehicle score and box offsets at every BEV location/anchor. **ROI** means *Region of Interest*: the 6–21 BEV cells covered by that box.

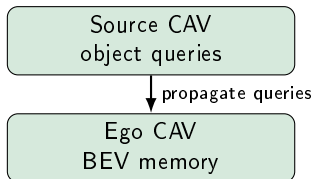
TrackFormer Training: Temporal Tracking → Spatial Association

Original TrackFormer



Frame $t - 1$ detections spawn queries that carry identity into frame t .

Our spatial adaptation



Source-CAV detections spawn queries that attend to ego-CAV BEV memory at the same timestamp.

Training objective from the TrackFormer paper. New objects use Hungarian matching; persistent tracks are matched by their known identity:

$$\hat{\sigma} = \arg \min_{\sigma} \sum_i \left[-\lambda_{\text{cls}} \hat{p}_{\sigma(i)}(c_i) + C_{\text{box}}(b_i, \hat{b}_{\sigma(i)}) \right], \quad C_{\text{box}} = \lambda_1 \|b_i - \hat{b}_{\sigma(i)}\|_1 + \lambda_{\text{giou}} C_{\text{giou}}.$$

$$\mathcal{L}_{\text{MOT}} = \sum_j \left[\text{CE}(c_j, \hat{p}_j) + \mathbb{1}_{j \in \pi} (\lambda_1 \mathcal{L}_1 + \lambda_{\text{giou}} \mathcal{L}_{\text{giou}}) \right], \quad c_j = 0 \text{ for unmatched queries.}$$

Why this can work: source queries attend to ego BEV memory; shared-frame geometry disambiguates similar vehicles. False negative and false positive queries teach recovery and rejection.

Exact Cross-Agent TrackFormer Pipeline

For source agent i : $F_i \in \mathbb{R}^{384 \times 100 \times 176}$ is its local BEV map and $P_i = \{(b_{i,k}^{\text{local}}, s_{i,k})\}$ are its local detected boxes and confidence scores.

Definitions: one proposal k from source agent i

- ▶ $\mathcal{R}_{i,k}$: **index set** of BEV cells covered by the rotated detected box $b_{i,k}$; it is not a feature vector.
- ▶ $X_{i,k} = [F_i(r_1); \dots; F_i(r_m)] \in \mathbb{R}^{m \times 384}$: the actual **ROI token data** selected from F_i using $\mathcal{R}_{i,k}$, with $m \approx 6\text{--}21$ cells.
- ▶ $g_{i,k} \in \mathbb{R}^9$: the normalized ego-frame geometry and confidence $[x_e, y_e, z_e, h, w, l, \sin \psi, \cos \psi, s_{i,k}]$.
- ▶ $h_{i,k} \in \mathbb{R}^{256}$: learned final decoder **query state**; an internal float vector, not a box or raw LiDAR data.
- ▶ $z_{i,k} \in \mathbb{R}^{128}$: the L2-normalized **association embedding** computed from $h_{i,k}$ and $g_{i,k}$.

Process and communication

Source: $F_i, P_i \rightarrow \mathcal{R}_{i,k}, X_{i,k}, g_{i,k} \rightarrow \text{TrackFormer} \rightarrow (h_{i,k}, z_{i,k})$; send selected query states, ego-frame boxes, scores, and embeddings—not the full F_i map.

Ego: builds $X_{e,k}, g_{e,k}$, propagates received $h_{i,k}$ through its BEV memory, then associates source and ego proposals. The detection head defines each ROI and explicitly conditions TrackFormer through $g_{i,k}$.

Embedding Results and Current Conclusion

Model / evaluation	Top-1 association accuracy
Single-cell MLP, held-out cache	≈ 0.21
Spatial TrackFormer + geometry head, clean validation	0.915
Spatial TrackFormer + geometry head, clean OPV2V test	0.891
Same clean model, noisy validation	0.907
Same clean model, noisy OPV2V test	0.891

Takeaway

Using all vehicle-covered BEV cells and spatial TrackFormer query propagation substantially improves cross-agent identity association. The embedding model is strong in isolation; the remaining work is to translate this association quality into better end-to-end fusion recall under noisy geometry.



T. Meinhardt et al., “TrackFormer: Multi-Object Tracking with Transformers,” CVPR, 2022.